2.1 What "Eventually" Means

WORKSHEET

The question

STEP III 2015, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q2

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself The definition contains the phrase "there exists a positive integer m". To prove that $(s_n) \leq (t_n)$, what exactly do you have to produce? To prove that it is *false*, what do you have to show?

STAGE 1 A linear function against a quadratic

Covers: Part (i)

Ask yourself The first terms do not matter. For which n is $1000n \leq n^2$? Divide both sides by n, which is positive.

STAGE 2 Is “eventually smaller” like ordinary “smaller”?

Covers: Part (ii)

Ask yourself Write out “it is *not* the case that $(s_n) \leq (t_n)$” in full, without the word “not” in front. How often must $s_n > t_n$ happen?

Pause and try Look for two sequences of positive numbers that swap places for ever, so that neither is eventually smaller than the other.

STAGE 3 Chaining

Covers: Part (iii)

Ask yourself Each hypothesis comes with its own m, and they may be different. Call them m_1 and m_2.

Pause and try Which single value of m makes both inequalities hold for every $n \geq m$?

STAGE 4 A quadratic against an exponential

Covers: Part (iv)

Ask yourself Make a table of n^2 and 2^n for $n = 1, \dots, 8$. From where does $n^2 \le 2^n$ seem to hold?

Pause and try Prove it by induction. To get from $n^2 \le 2^n$ to $(n+1)^2 \le 2 \cdot 2^n$, it is enough to know that $(n+1)^2 \le 2n^2$. For which n is that true?

What's really going on

The language of analysis

For about 150 years after Newton and Leibniz, mathematicians used limits and infinitesimals brilliantly but could not say exactly what they meant. Bishop Berkeley mocked infinitesimals in 1734 as "the ghosts of departed quantities", and he had a point. The repair came in the nineteenth century, when Augustin-Louis Cauchy and then Karl Weierstrass replaced vague talk of "becoming infinitely close" with statements built from "for every" and "there exists". The modern definition

$$a_n \to L \iff \text{for every } \varepsilon > 0 \text{ there exists } N \text{ such that } |a_n - L| < \varepsilon \text{ for all } n \ge N$$

has exactly the shape of this question's definition, with one extra quantifier at the front. It is the first thing you will meet in a university analysis course.

Orders of growth

Part (iv) is one small piece of a famous ladder. For any fixed $k > 0$ and $c > 1$,

$$(\ln n) \leq (n^k), \qquad (n^k) \leq (c^n), \qquad (c^n) \leq (n!), \qquad (n!) \leq (n^n),$$

and in fact each sequence is not just eventually smaller than the next but eventually smaller than *any constant multiple* of it. G. H. Hardy wrote a whole book, *Orders of Infinity* (1910), about this kind of comparison. Today it is the everyday language of computer science, where "this algorithm takes $O(n^2)$ steps" means that the number of steps is eventually at most a constant times n^2. The difference between n^2 and 2^n is the difference between a program that finishes and one that will not finish before the Sun burns out.

Orders that are not total

Part (ii) shows that "eventually smaller" is not a *total order*: some pairs of sequences cannot be compared at all. Relations like this, transitive but not comparing everything, are called *preorders* or *partial orders*, and they turn up everywhere: divisibility of integers, inclusion of sets, and "is a subgroup of" are all examples. Recognising which properties a relation has, and which it doesn't, is part of the training of a mathematician.

Try this next

1. **The specimen warm-up.** Try STEP III Specimen paper, Question 2 (step.maths.org/questions/spec-s3-q2), from the paper issued before the first STEP in 1987. (*Hint for the last part:* use the next term of the binomial expansion.)

2. **Both ways round.** Prove that if $(s_n) \leq (t_n)$ and $(t_n) \leq (s_n)$, then $s_n = t_n$ for all sufficiently large n.

3. **Racing to infinity.** Find two *increasing* sequences of positive numbers, both tending to infinity, neither of which is eventually smaller than the other.

4. **Up the ladder.** Show that $(2^n) \leq (n!)$, giving an explicit m. Then show that $(n^{10}) \leq (2^n)$ and find the smallest m that works. (A spreadsheet helps for the second part; the answer is surprisingly large.)

Further reading

- **Lara Alcock, *How to Study for a Mathematics Degree*** (Oxford University Press). Chapter 3 is about exactly this kind of logical language and why new undergraduates find it hard.
- **Lara Alcock, *How to Think About Analysis*** (Oxford University Press). A friendly guide to the definitions of first-year analysis.

- **Daniel Velleman, *How to Prove It*** (Cambridge University Press). A systematic introduction to logic, quantifiers and proof.

The full guided solution is in Section 2.1:
https://beyondstep.keithmcnulty.org/ch02/eventually.html